

Erdős Problem #711

EP-711

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $f(n, m)$ be minimal such that in $(m, m + f(n, m))$ there exist distinct integers $a_1, \dots, a_n$ such that $k \mid a_k$ for all $1 \leq k \leq n$. Prove that $\max_m f(n, m) \leq n^{1+o(1)}$ and that $\max_m (f(n, m) - f(n, n)) \rightarrow \infty$.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>